LADDER OF LIFE DETECTION

NEVEU ET AL. (2018)

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RUNG	FEATURE	MEASUREMENT	TARGET	LIKELIHOOD	INSTRUMENTAL CRITERIA			CONTEXTUAL CRITERIA				
Roughly, subjectively ordered by (top to bottom): 1. decreasing strength of evidence for life 2. increasing ease of measurement	Listed in no specific order within a given rung			…that the feature would be a biosignature, given the criteria to the right	QUANTIFIABLE Detectability	CONTAMINATION-FREE Likelihood of false positive	REPEATABLE	DETECTABLE Detectability	SURVIVABLE Likelihood of false negative	RELIABLE Ambiguity of feature	COMPATIBLE Specificity to Earth life	LAST-RESORT Ambiguity of interpretation
DARWINIAN EVOLUTION	Changes in inheritable traits in response to selective pressures	Not practical under mission constraints	In situ Sample return	No	—	—	Number of replicates depends on: Characteristics of the instrument. Heterogeneity of the sample. Likelihood of systematic errors. Required values of the relevant statistical parameters. Value and cost of information.	—	N/A (extant)	—	—	—
GROWTH AND REPRODUCTION	Concurrent life stages or identifiable reproductive form, motility	Cell(like?) structures in multiple stages	In situ Sample return	Low	Hard	Low		Med (don't identify stages, timing off, sample size low)	High?	Ambiguous. What is a cell? What morphological differences exist?	Earth	Med/High
METABOLISM	Major element or isotope fractionations indicative of metabolism	Deviation from abiotic fractionation controlled by thermodynamic equilibrium and/or kinetics	Remote sensing In situ Sample return	Low/Med	Easy	High		Medium	High	Hinges on understanding or context	Earth?	Low
	Response to substrate addition	Waste output (compound, heat)	In situ Sample return	Low/Med	Easy	Low		High	N/A (extant)	Hinges on understanding context	Earth?	Medium
	Co-located reductant and oxidant	Deviation from abiotic distribution controlled by thermodynamic equilibrium and/or kinetics	Remote sensing In situ Sample return	Med/High	Med (linked to specificity of instrument)	Low/Med		Med/High	High	Mixed reactions, large inventory of chemistries	Generic	Low/Med
MOLECULES & STRUCTURES CONFERRING FUNCTION	Polymers that support information storage and transfer for terran life (DNA, RNA)	Abundance	In situ Sample return	Low	Hard (instrument specificity must be high); RNA hard to measure on Earth	DNA: high; RNA: low (reactive)		Low (technology limited, only terran); RNA highly reactive	Low (hydrolysis in water)	Reliable	Earth	Negligible
	Structural preferences in organic molecules (non-random and enhancing function)	Polymer with repeating charge	In situ Sample return	Low/Med	Need a lot of material and overprinting must be discernable	Low		Med/High	Low (hydrolysis in water, diagenesis)	How much preference needed to detect?	Generic	Low
		Enantiomeric excess >20% in multiple amino acid types	In situ Sample return	High	How much excess necessary?	Low		Low	Medium	Mixed sample both processes present	Generic	Low
	Pigments as evidence of non-random chemistries (e.g. specific pathways)	Spectral feature and/or color, otherwise see "structural preferences"	Remote sensing In situ Sample return	Low/Med	Easy (fluorescence)	Low		Low (Limitation of what we are looking for)	Low (diagenesis)	How to define pigment as we don't know it?	Earth (can one abstract?)	Very low
POTENTIAL BIOMOLECULE COMPONENTS	Organics not found abiotically (e.g. hopanes, ATP, histidine)	Presence	In situ Sample return	Medium	Easy if enough material	Low		High	High	Low	Earth?	High
	Complex organics (e.g. nucleic acid oligomers, peptides, PAH)	Presence	Remote sensing (PAH) In situ Sample return	High	Easy if enough material	Low		High	Medium	Abiotic production known	Generic	Med/High
	Monomeric units of biopolymers (nucleobases, amino acids, lipids for compartmentalization)	Presence	Remote sensing In situ Sample return	Med/High	Limit of detection, need a lot of material	High		High	Med (diagenesis)	Abiotic pathways known	Generic	Medium
POTENTIAL METABOLIC BYPRODUCTS	Distribution of metals e.g. V in oil of Fe, Ni, Mo/W, Co, S, Se, P	Presence	In situ Sample return	Medium	Easy except background issue	Low		High	High	Background known	Generic	Medium
	Patterns of complexity (organics)	Deviation from equilibrium (P(Poisson distribution of pathway complexity) <0.01?) or abiotic kinetic distribution	In situ Sample return	High	Background issue, material limited	Low		High	Medium?	Limited documentation of abiotic vs. biotic differences	Generic	Medium
BIOFABRICS	Textures	Biologically mediated morphologies, preferably with co-located composition	In situ Sample return	Medium	Medium	Low		Medium	High?	Highly ambiguous	Earth	High
HABITABILITY	Liquid water, building blocks, energy source, gradients. Redox, temperature, pH, energy, disequilibria.											

Rows indicate features of life; columns indicate criteria of Table 1. Features highlighted in green indicate evidence for extant life. Edits to the Ladder are encouraged; see Neveu et al. (2018), doi: 10.1089/ast.2017.1773. • **Likelihood** — A summary of the factors to the right including specificity, ambiguity, false positive, false negative and detectability. • **False Positive** — A positive determination based on detection due to contamination or a positive detection caused by an unintended interference with the method of detection. • **False Negative** — Lack of detection when the factor is actually present either in the sample or the system at large. For example, low signal to background or signal below the limit of detection.

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